Academic Year: 2025 - 2026

Bachelor in Mathematics, option: Data Science

Fraud Detection Using Machine Learning

STUDENT

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Fraud detection is an important challenge in financial systems because fraudulent transactions can cause financial loss and reduce customer trust.

The main problem of this project is to classify each transaction as either:

- Non-fraudulent transaction
- Fraudulent transaction

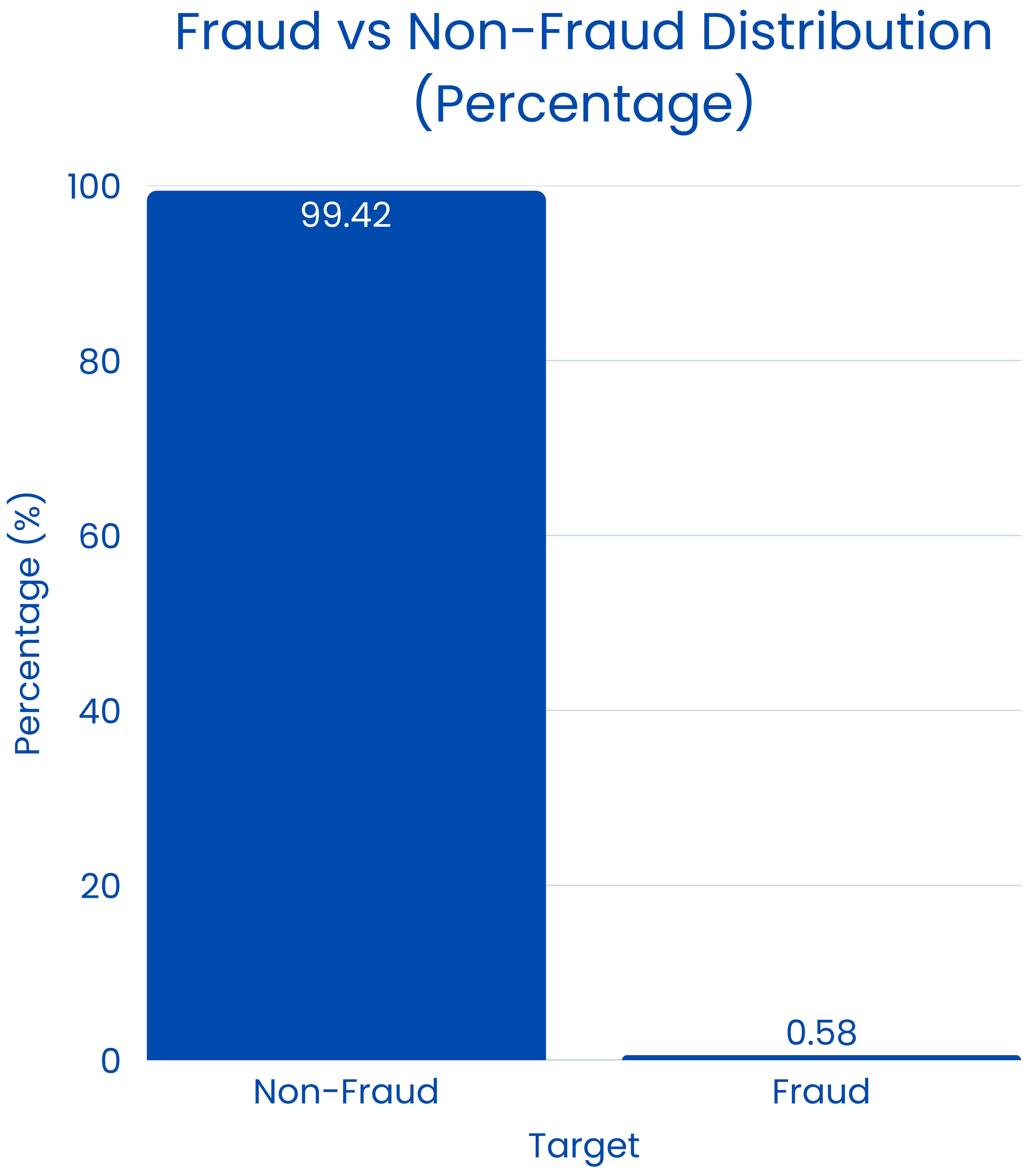
A major challenge is that fraud cases are rare compared to normal transactions, making the dataset highly imbalanced.

- Build machine learning models for fraud detection
- Compare supervised and unsupervised approaches
- Evaluate models using suitable metrics for imbalanced data
- Apply PCA and analyze its effect on performance
- Identify the most important fraud-related features
- Compare models from a business cost perspective

The dataset used in this project was obtained from Kaggle and contains credit card transaction records.

Key information:

- More than 1 million transactions
- Target variable: is_fraud
- 0 = Non-fraud
- 1 = Fraud
- Includes transaction, customer, merchant, category, time, and location information



Before training the models, the data was prepared using several preprocessing steps:

- Removed unnecessary columns
- Handled missing values using imputation
- Scaled numerical features
- Encoded categorical features using One-Hot Encoding
- Created new features: transaction hour, day, weekday and customer age

The project followed a complete machine learning workflow:

- Load and explore the dataset
- Perform preprocessing and feature engineering
- Split the data into training and testing sets
- Train multiple fraud detection models
- Evaluate model performance
- Apply cross-validation
- Apply PCA and compare results
- Analyze feature importance
- Perform business cost evaluation

Supervised models: Logistic Regression, Random Forest and XGBoost

Unsupervised anomaly detection models: Isolation Forest and One-Class SVM

Supervised models learn from labeled fraud data, while unsupervised models detect unusual transactions.

Since the dataset is highly imbalanced, accuracy alone is not reliable.

The models were evaluated using:

Precision

Correct fraud predictions

Recall

Fraud cases detected

F1-score

Balance between precision and recall

ROC-AUC

Class separation ability

Confusion Matrix

Prediction breakdown

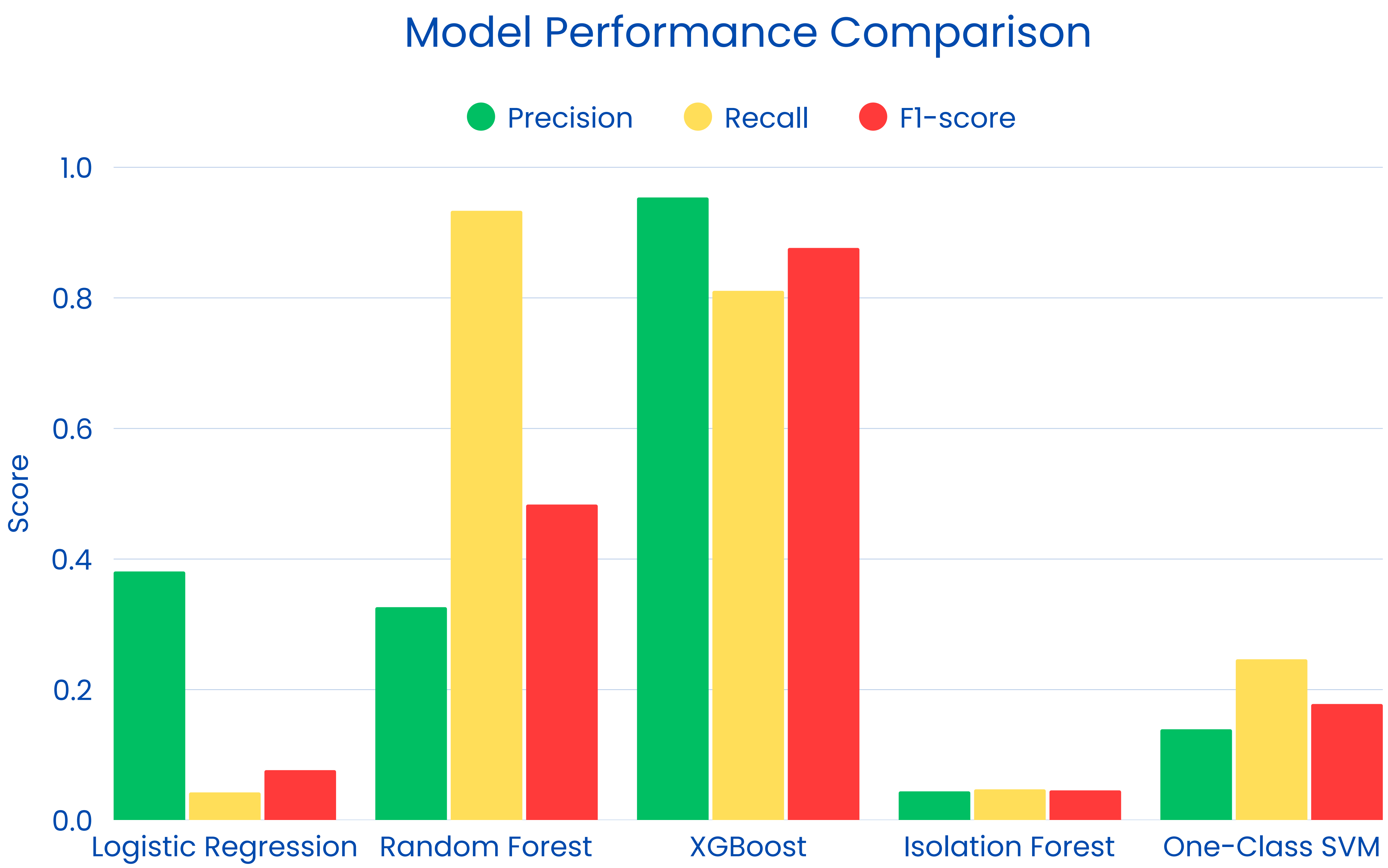
In fraud detection, recall is especially important because missing fraud can be costly.

The model comparison showed that:

- Logistic Regression had very low recall and missed most fraud cases.
- Isolation Forest and One-Class SVM performed weakly.
- Random Forest achieved very high recall but produced many false positives.
- XGBoost achieved the best balance between precision, recall, and F1-score.

Main finding:

XGBoost was the best overall model basec on machine learning metrics.



Model	PCA	Recall	F1-score	ROC-AUC
Logistic Regression	No	0.0426	0.0767	0.8372
Logistic Regression	Yes	0.0460	0.0825	0.8368
XGBoost	No	0.8108	0.8765	0.9955
XGBoost	Yes	0.6789	0.7743	0.9908

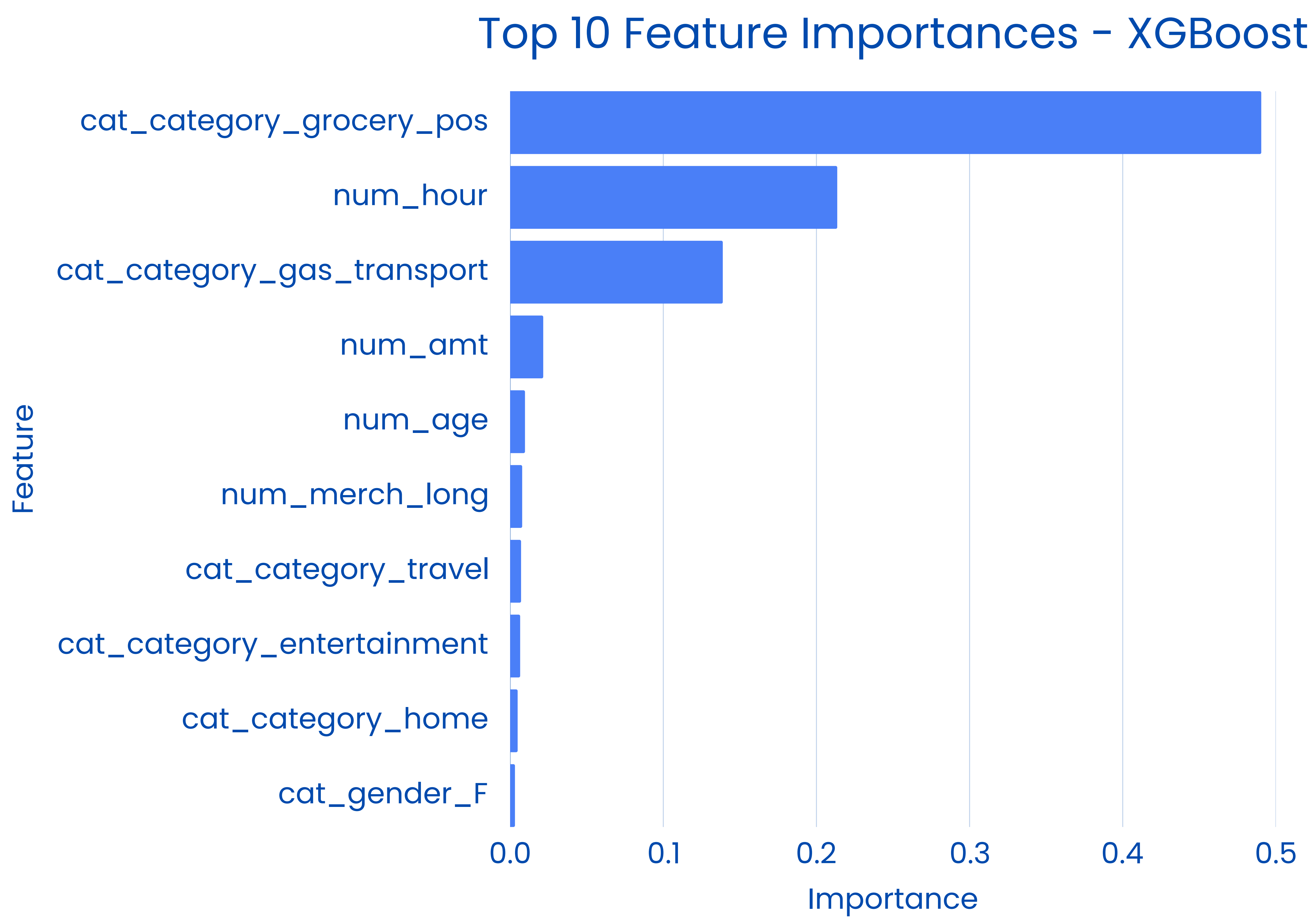
5-fold Stratified Cross-Validation was used to test model stability.

Main findings:

- Logistic Regression remained weak
- XGBoost was stable and reliable
- PCA did not improve performance
- PCA reduced XGBoost recall and F1-score

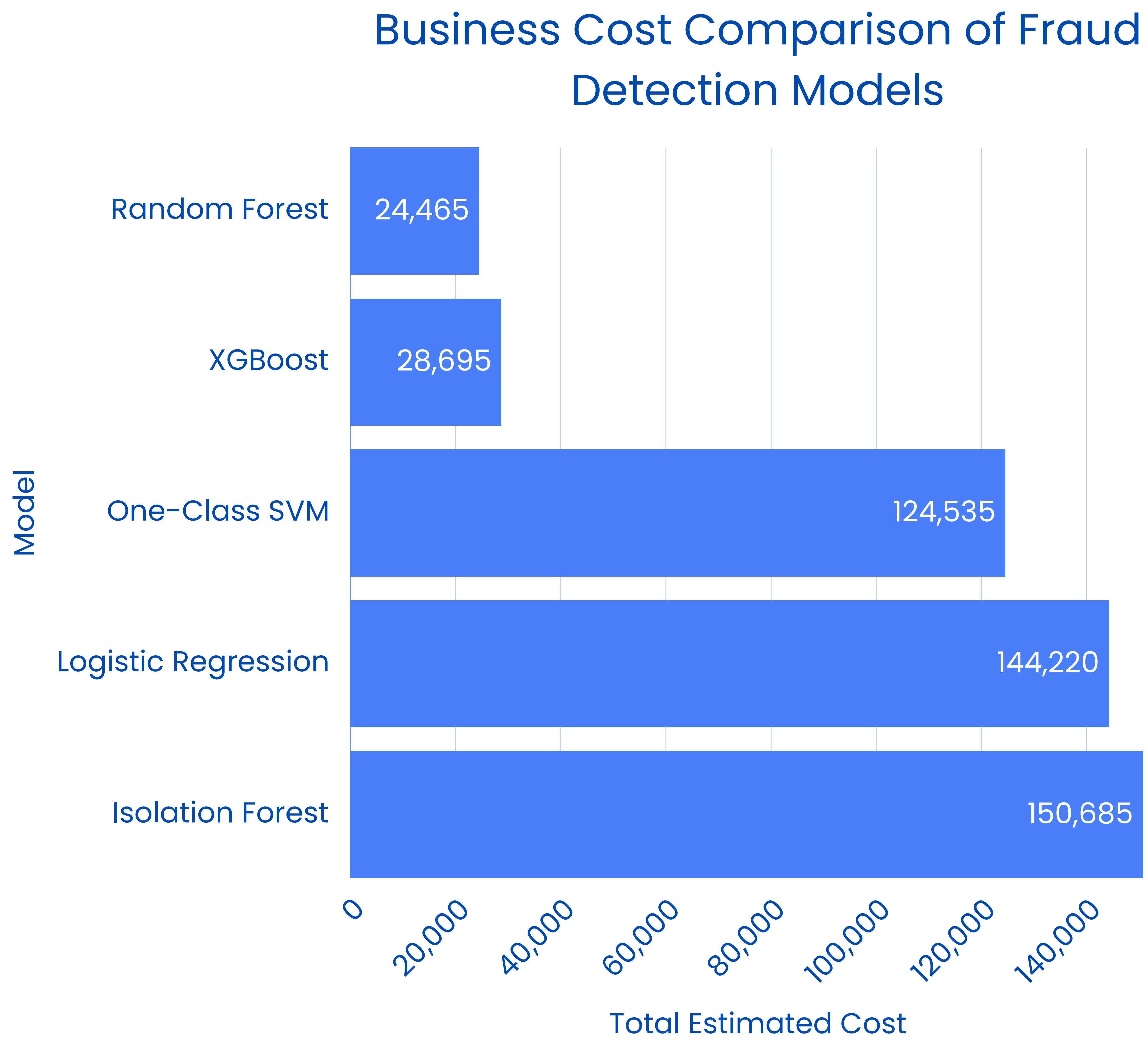
Conclusion:

The original feature space was more useful for fraud detection.



Main findings:

- Transaction category was the most important factor
- Transaction hour was highly influential
- Fraud depends more on behavior patterns than amount alone



$$\text{Total Cost} = FN \times CostFN + FP \times CostFP$$

Main findings:

- XGBoost was best statistically
- Random Forest had the lowest business cost
- Random Forest missed fewer fraud cases

Fraud detection is not only about model accuracy; it is about choosing the model that matches the real-world objective.

Technical Performance

Best model: XGBoost

- Best balance between precision and recall
- Highest F1-score
- Stable with cross-validation

Business Perspective

Best model: Random Forest

- Lowest estimated business cost
- Missed fewer fraud cases
- Useful when fraud loss is the priority

Final Insight

Main lesson: Objective matters

- PCA did not improve results
- Category and time were key indicators
- Supervised models performed best

Thank You!